

Chomsky Normal Form

Example1

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow aX \mid Yb \\ X &\rightarrow S \mid /\backslash \\ Y &\rightarrow bY \mid b \end{aligned}$$

Solution 1

Step 1 - Kill all $/\backslash$ productions:

By inspection, the only nullable nonterminal is X .
Delete all $/\backslash$ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without $/\backslash$ productions, is:

$$\begin{aligned} S &\rightarrow aX \mid a \mid Yb \\ X &\rightarrow S \\ Y &\rightarrow bY \mid b \end{aligned}$$

Step 2 - Kill all unit productions:

The only unit production is $X \rightarrow S$, where the S can be replaced with all S 's non-unit productions (i.e. aX , a , and Yb).

The new CFG, without unit productions, is:

$$\begin{aligned} S &\rightarrow aX \mid a \mid Yb \\ X &\rightarrow aX \mid a \mid Yb \\ Y &\rightarrow bY \mid b \end{aligned}$$

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$$\begin{aligned} S &\rightarrow AX \mid YB \mid a \\ X &\rightarrow AX \mid YB \mid a \\ Y &\rightarrow BY \mid b \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Step 4 - Shorten the strings of nonterminals to length 2.

All nonterminal strings on the RHS in the above CFG are already the required length, so the CFG is in CNF.

Example2

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow AA \\ A &\rightarrow B \mid BB \\ B &\rightarrow abB \mid b \mid bb \end{aligned}$$

Solution 2

Step 1 - Kill all ϵ productions:

There are no ϵ productions, so none of the non-terminals is nullable.
The CFG remains unchanged.

Step 2 - Kill all unit productions:

The only unit production is $A \rightarrow B$, where the B can be replaced with all B 's non-unit productions (i.e. all of them).

The new CFG, without unit productions, is:

$$\begin{aligned} S &\rightarrow AA \\ A &\rightarrow BB \mid abB \mid b \mid bb \\ B &\rightarrow abB \mid b \mid bb \end{aligned}$$

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$$\begin{aligned} S &\rightarrow AA \\ A &\rightarrow BB \mid XYB \mid b \mid YY \\ B &\rightarrow XYB \mid b \mid YY \\ X &\rightarrow a \\ Y &\rightarrow b \end{aligned}$$

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

$$\begin{aligned} S &\rightarrow AA \\ A &\rightarrow BB \mid RB \mid b \mid YY \\ B &\rightarrow RB \mid b \mid YY \\ R &\rightarrow XY \\ X &\rightarrow a \\ Y &\rightarrow b \end{aligned}$$

Example3

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow XaX \mid bX \mid Y \\ X &\rightarrow XaX \mid XbX \mid \epsilon \\ Y &\rightarrow ab \end{aligned}$$

Solution 3

Step 1 - Kill all ϵ productions:

By inspection, the only nullable nonterminal is X .
Delete all ϵ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without ϵ productions, is:

$$\begin{aligned} S &\rightarrow XaX \mid aX \mid Xa \mid a \mid bX \mid b \mid Y \\ X &\rightarrow XaX \mid aX \mid Xa \mid a \mid XbX \mid bX \mid Xb \mid b \\ Y &\rightarrow ab \end{aligned}$$

Step 2 - Kill all unit productions:

The only unit production is $S \rightarrow Y$, where the Y can be replaced with all Y 's non-unit productions (i.e. ab). Furthermore, the Y -production can be completely removed, since its only purpose is to turn S into ab in one particular production sequence: $S \rightarrow Y \rightarrow ab$.

The new CFG, without unit productions, is:

$$\begin{aligned} S &\rightarrow XaX \mid aX \mid Xa \mid a \mid bX \mid b \mid ab \\ X &\rightarrow XaX \mid aX \mid Xa \mid a \mid XbX \mid bX \mid Xb \mid b \end{aligned}$$

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$$\begin{aligned} S &\rightarrow XAX \mid AX \mid XA \mid BX \mid AB \mid a \mid b \\ X &\rightarrow XAX \mid AX \mid XA \mid XBX \mid BX \mid XB \mid a \mid b \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

$$\begin{aligned} S &\rightarrow XR \mid AX \mid XA \mid BX \mid AB \mid a \mid b \\ R &\rightarrow AX \\ X &\rightarrow XR \mid AX \mid XA \mid XQ \mid BX \mid XB \mid a \mid b \\ Q &\rightarrow BX \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Example4

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow SS \mid A \mid B \\ A &\rightarrow SS \mid AS \mid a \\ B &\rightarrow \epsilon \end{aligned}$$

Solution 4

Step 1 - Kill all ϵ productions:

By inspection, the nullable nonterminals are S, A, and B.
Delete all ϵ productions and add new productions, with all possible combinations of the nullable nonterminals removed.

The new CFG, without ϵ productions, is:

$$\begin{aligned} S &\rightarrow SS \mid A \mid S \\ A &\rightarrow SS \mid AS \mid S \mid A \mid a \end{aligned}$$

Step 2 - Kill all unit productions:

The unit productions $S \rightarrow S$ and $A \rightarrow A$ are redundant and can be removed.
The only unit production for S is then $S \rightarrow A$, which can be replaced with all A's non-unit productions. The only unit production for A is then $A \rightarrow S$, which can be replaced with all S's non-unit productions.

The new CFG, without unit productions, is:

$$\begin{aligned} S &\rightarrow SS \mid AS \mid a \\ A &\rightarrow SS \mid AS \mid a \end{aligned}$$

Step 3 - Replace all mixed strings with solid nonterminals.

Step 4 - Shorten the strings of nonterminals to length 2.

Steps 3 and 4 are not necessary, since the CFG is already in CNF.
The CFG can however be shortened to:

$$S \rightarrow SS \mid a$$

Example5

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow XaX \mid YY \mid XY \\ X &\rightarrow \epsilon \mid b \\ Y &\rightarrow Xa \end{aligned}$$

Solution 5

Step 1 - Kill all ϵ productions:

By inspection, the only nullable nonterminal is X.
Delete all ϵ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without ϵ productions, is:

$$\begin{aligned} S &\rightarrow XaX \mid aX \mid Xa \mid a \mid YY \mid XY \mid Y \\ X &\rightarrow b \\ Y &\rightarrow Xa \mid a \end{aligned}$$

Step 2 - Kill all unit productions:

The only unit production is $S \rightarrow Y$, where the Y can be replaced with all Y's non-unit productions (i.e. all of them).



The new CFG, without unit productions, is:

$$\begin{aligned} S &\rightarrow XaX \mid aX \mid \mathbf{Xa} \mid \mathbf{a} \mid YY \mid XY \\ X &\rightarrow b \\ Y &\rightarrow Xa \mid a \end{aligned}$$

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$$\begin{aligned} S &\rightarrow XAX \mid AX \mid XA \mid YY \mid XY \mid a \\ X &\rightarrow b \\ Y &\rightarrow XA \mid a \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

$$\begin{aligned} S &\rightarrow XR \mid AX \mid XA \mid YY \mid XY \mid a \\ R &\rightarrow AX \\ X &\rightarrow b \\ Y &\rightarrow XA \mid a \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Example6

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow aX \mid bS \mid ab \mid X \\ X &\rightarrow aX \mid a \mid /\backslash \end{aligned}$$

Solution 6

Step 1 - Kill all $/\backslash$ productions:

Both S and X are nullable nonterminals.

Delete all $/\backslash$ productions and add new productions, with all possible combinations of the nullable S and X removed.

The new CFG, without $/\backslash$ productions, is:

$$\begin{aligned} S &\rightarrow aX \mid bS \mid ab \mid X \mid a \mid b \\ X &\rightarrow aX \mid a \end{aligned}$$

Step 2 - Kill all unit productions:

The only unit production is $S \rightarrow X$, where the X can be replaced with all X's non-unit productions (i.e. aX and a).

The new CFG, without unit productions, is:



WikiStudent
for Unisa students

<http://wikistudent.ws/Unisa>

$S \rightarrow aX \mid bS \mid ab \mid a \mid b$
 $X \rightarrow aX \mid a$

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$S \rightarrow AX \mid BS \mid AB \mid a \mid b$
 $X \rightarrow AX \mid a$
 $A \rightarrow a$
 $B \rightarrow b$

Step 4 - Shorten the strings of nonterminals to length 2.

All nonterminal strings on the RHS in the above CFG are already the required length, so the CFG is in CNF.

Question 7

Convert the following CFG to Chomsky Normal Form (CNF):

$S \rightarrow aX \mid Y \mid bab$
 $X \rightarrow /\backslash \mid Y$
 $Y \rightarrow bb \mid bXb$

Solution 7

Step 1 - Kill all $/\backslash$ productions:

By inspection, the nullable nonterminal is X.

Delete all $/\backslash$ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without $/\backslash$ productions, is:

$S \rightarrow aX \mid a \mid Y \mid bab$
 $X \rightarrow Y$
 $Y \rightarrow bb \mid bXb$

Step 2 - Kill all unit productions:

The only unit productions are $S \rightarrow Y$ and $X \rightarrow Y$, where the Y in both cases can be replaced with all Y's non-unit productions (i.e. bb and bXb).

The new CFG, without unit productions, is:

$S \rightarrow aX \mid a \mid bab \mid bb \mid bXb$
 $X \rightarrow bb \mid bXb$

(The Y productions are no longer necessary in the CFG).

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$$\begin{aligned} S &\rightarrow AX \mid a \mid BAB \mid BB \mid BXB \\ X &\rightarrow BB \mid BXB \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

$$\begin{aligned} S &\rightarrow AX \mid BQ \mid BB \mid BR \mid a \\ X &\rightarrow BB \mid BR \\ Q &\rightarrow AB \\ R &\rightarrow XB \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Example8

Convert the following CFG to Chomsky Normal Form (CNF):

$$\begin{aligned} S &\rightarrow aY \mid Ybb \mid Y \\ X &\rightarrow /\backslash \mid a \\ Y &\rightarrow aXY \mid bb \mid XXa \end{aligned}$$

Solution 8

Step 1 - Kill all $/\backslash$ productions:

By inspection, the only nullable nonterminal is X.
Delete all $/\backslash$ productions and add new productions, with all possible combinations of the nullable X removed.

The new CFG, without $/\backslash$ productions, is:

$$\begin{aligned} S &\rightarrow aY \mid Ybb \mid Y \\ X &\rightarrow a \\ Y &\rightarrow aXY \mid aY \mid bb \mid XXa \mid Xa \mid a \end{aligned}$$

Step 2 - Kill all unit productions:

The only unit production is $S \rightarrow Y$, where the Y can be replaced with all Y's non-unit productions (i.e. all of them).

The new CFG, without unit productions, is:

$$\begin{aligned} S &\rightarrow aY \mid Ybb \mid aXY \mid aY \mid bb \mid XXa \mid Xa \mid a \\ X &\rightarrow a \\ Y &\rightarrow aXY \mid aY \mid bb \mid XXa \mid Xa \mid a \end{aligned}$$

Step 3 - Replace all mixed strings with solid nonterminals.

Create extra productions that produce one terminal, when doing the replacement.

The new CFG, with a RHS consisting of only solid nonterminals or one terminal is:

$$\begin{aligned} S &\rightarrow AY \mid YBB \mid AX Y \mid AY \mid BB \mid XXA \mid XA \mid a \\ X &\rightarrow a \\ Y &\rightarrow AX Y \mid AY \mid BB \mid XXA \mid XA \mid a \\ A &\rightarrow a \\ B &\rightarrow b \end{aligned}$$

Step 4 - Shorten the strings of nonterminals to length 2.

Create new, intermediate nonterminals to accomplish this.

The new CFG, in CNF is:

$$\begin{aligned} S &\rightarrow AY \mid YR \mid AQ \mid AY \mid BB \mid XP \mid XA \mid a \\ X &\rightarrow a \\ Y &\rightarrow AQ \mid AY \mid BB \mid XP \mid XA \mid a \\ A &\rightarrow a \\ B &\rightarrow b \\ R &\rightarrow BB \\ Q &\rightarrow XY \\ P &\rightarrow XA \end{aligned}$$